# MOAIN AL-ABBASI

**Software Engineer | Full-Stack Systems · AI Agents · Cybersecurity**

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## SUMMARY

Software engineer who designs, builds and ships complete business systems on my own, from database to user interface to deployment. I have delivered POS/ERP, attendance, notification and remote-access platforms, offline-first mobile apps, multi-agent AI systems and fast, SEO-focused websites for clients in Saudi Arabia and Yemen. I write tests for every system, document my decisions, and hand clients full ownership of the source code. I work fully remote in the same time zone as Saudi Arabia.

## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript, Python, Go, Dart, Kotlin, SQL, PL/SQL, PHP
**Backend:** NestJS, Node.js, FastAPI, Hono, Express, Prisma, Drizzle, REST APIs, BullMQ, Redis, WebSockets
**Frontend:** React 19, Next.js, Angular 22, Tailwind CSS, PWA, offline-first (IndexedDB), Arabic RTL, accessibility (WCAG 2.1 AA)
**Mobile:** Flutter (Android), Kotlin with Jetpack Compose, barcode scanning (ML Kit), Bluetooth thermal printing
**Databases:** PostgreSQL, Oracle, MySQL, SQLite; data migration from legacy and closed systems
**AI & Agents:** LLM integration (OpenAI, Anthropic, Gemini), LangGraph, LangChain, multi-agent orchestration, RAG and hybrid search (FTS5, vectors, BM25, RRF), Model Context Protocol (MCP) servers, OCR, Arabic NLP
**Security:** SOC operations with Wazuh SIEM, Suricata IDS, YARA, MITRE ATT&CK; digital forensics (Autopsy, FTK Imager); Zero-Trust remote access
**DevOps & Quality:** Docker, GitHub Actions CI/CD, Cloudflare (Pages, Workers, R2), Vercel, Linux VPS; pytest, Vitest, Playwright, mypy (strict)
**Web Growth:** technical SEO, Core Web Vitals, Google Search Console, Google Ads API, conversion optimisation, Saudi PDPL compliance

## PROFESSIONAL EXPERIENCE

**Founder & Lead Software Engineer** · Alabbasi Tech, software house · Sana'a, Yemen (Remote) 2026 – Present

I build custom systems that replace closed, per-seat software. Every client receives the full source code, Arabic documentation and three months of support.

- Built a **POS / ERP system** to replace a paid closed product (YemenSoft Onyx Pro), covering inventory, purchasing, accounting and cashier, and migrated the client's existing data. It prints Arabic receipts on Bixolon thermal printers over Bluetooth. NestJS, React 19, Oracle PL/SQL, Vitest.
- Built **Nexus Notify**, a multi-tenant, multi-channel notification platform with queues, real-time delivery reports, an Angular dashboard and a native Android SMS gateway with dual-SIM support. NestJS, PostgreSQL, Prisma, BullMQ, Redis, Kotlin.
- Replaced vendor attendance software with an **enterprise attendance system** that reads ZKTeco iFace950 devices directly, combines all branches into one system and exports payroll-ready reports. Go, PostgreSQL.
- Built a **Zero-Trust remote-access platform** with digitally signed agents, per-user policies and an audit log for every session, plus a single-command cross-platform agent. Go, SSH, NetBird.
- Delivered an **offline-first event and hospitality management PWA** covering events, staff, inventory, invoices and statements, which keeps working at venues without internet. Angular 22, IndexedDB.
- Modernising a **legacy PHP/MySQL accounting system** by building a new NestJS / Prisma / React layer around it step by step, so operations never stop for a rewrite.

**Web & Growth Engineer** · Saudi hospitality businesses · Jeddah, Saudi Arabia (Remote) 2026 – Present

- Built and maintain production websites with technical SEO, conversion optimisation and PDPL compliance: keifaldiafa.com, asoulaldiafa.com, qahwajeen.com. Next.js, Hono.
- Built phase 1 of a **multi-tenant Google Ads management platform**: live integration with the official Google Ads API, a 12-table PostgreSQL schema and a reporting dashboard. Next.js 15, Prisma.
- Built an **AI content-publishing engine**: generates posts from a media archive, sends each one to a person for approval, then schedules publishing. It has a kill switch and runs in dry-run mode by default. Python, FastAPI, APScheduler.

## AI AGENTS & APPLIED AI

**Basira** · basirapp.site · source — Checks Quran and Hadith quotations by matching them exactly against 71,987 licensed source records. The AI model only proposes matches; a rule-based engine makes every decision. Results: 150/150 on the evaluation set, 0 false alarms out of 500 verbatim segments, 78.54% on IslamicEval 2025 (subtask 1B), 487 automated tests. Available as a REST API, an MCP server and a Telegram bot. Entry to the AI in Service of Islamic Content Challenge 2026. Python, FastAPI, React.

**Mansur AI** — Arabic multi-agent system in which planner, executor, verifier and learner agents are coordinated with LangGraph. It has 13 agents and 38 tools, picks the right model for each task with fallback and cost tracking, keeps layered memory, and saves successful runs as reusable skills. Arabic web interface with live streaming over WebSocket; 82 passing tests.

**Ai_Alabbasi** — Self-hosted autonomous coding agent. The model can be swapped in one line (cloud or local); it plans, runs Python in a sandbox, tests and fixes its own code, and uses hybrid retrieval (FTS5 plus vectors, merged with RRF).

**Moain Promoter AI** — Seven specialised agents under one orchestrator that write bilingual posts and publish them to six social platforms, with a Next.js dashboard. Python, Celery, Redis, Playwright.

**MCP & agent tooling** — Diafa Studio: produces Arabic PDF documents and exposes 18 MCP tools to external agents. A Telegram MCP server. Bina: reusable specialist agents, including a website audit agent with 205 checks.

## SELECTED PROJECTS

**Idhaat** · source — Live-tutoring platform. Zoom rooms are assigned automatically, with a private Jitsi stream secured by JWT when rooms run out. Recordings upload to Cloudflare R2 during class, carry a moving watermark and are deleted after 48 hours. Includes homework, messaging and full course finances. TypeScript, PWA.

**Grocery Ledger** · source — Offline Android accounting app and POS for small shops, with camera, USB or Bluetooth barcode scanning and append-only ledgers. 157 passing tests, v2.2.1. Flutter.

**Sijil** · source — Offline digital debt ledger for shops, usable by staff who cannot read. 93 passing tests. Flutter.

**S-ACM** · source — Academic content-management system with AI summarisation and question generation. React 19, Hono, PostgreSQL, Drizzle.

**Ghaida** · source — Household guidebook for domestic workers in their own language, with AI voice for every sentence. Offline PWA with an admin panel; all 27 API tests pass.

## EDUCATION

**Graduation Project: Security Operations Center (SOC)** · team project · 2026 · source
Open-source SOC built on Wazuh 4.14.7, Suricata, YARA, VirusTotal, auditd and MikroTik syslog, with an AI analyst whose actions require human approval. 14 use cases verified on a live lab, including automatic blocking of SSH brute-force attacks. Also built a hands-on digital forensics lab (Autopsy, FTK Imager, Windows Registry and event logs).

## LANGUAGES

Arabic: native · English: professional working proficiency